

Topic Modeling

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Topic Modeling

- An unsupervised Machine Learning Technique
- Automatically discovers hidden themes in a collection of documents
- Each document may have multiple themes
- The algorithm gives a probability to each theme
- Useful in summarizing large document and trend analysis

Topic Modeling

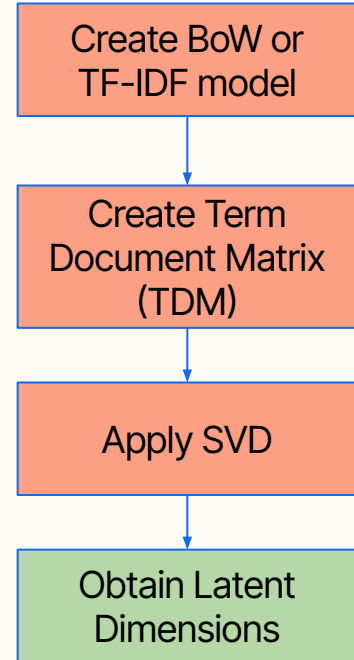
- Text must be converted into numerical form
- Most classical models use Bag-of-Words.
- Eg. Data science is fun $\rightarrow$ {data:1, science:1, fun:1}
- Modern models use Word Embeddings
(Word2Vec, GloVe) or Contextual Embeddings
(BERT, RoBERTa)

Clustering and Topic Modeling

Clustering	Topic Modeling
Each document belongs to one cluster	Each document is a mixture of topics
Hard Assignment	Soft Probability assignment
No such downstream use	Results can be used for clustering
Eg. Document Category: Politics	Eg. Document Themes: Politics-40%, Economics-30%, Policy:30%

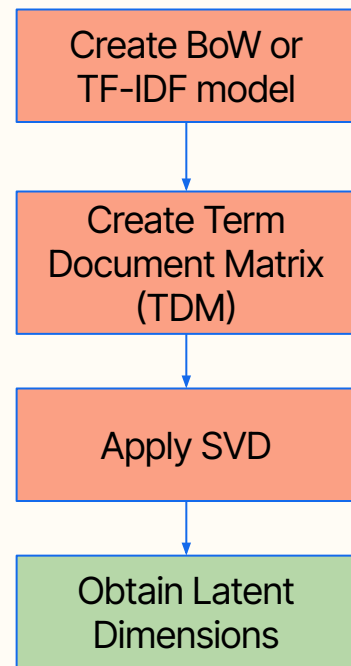
Latent Semantic Analysis

- In TDM, rows are words in vocabulary, columns are documents in corpus. Size of matrix is $V \times D$.
- Entry X_{ij} = Frequency of word i in document j .
- $X_{V \times D} = R_{V \times V} W_{V \times D} S_{D \times D}^T$
- After Truncation: $X \sim U_{V \times k} W_{k \times k} V_{k \times D}^T$



Latent Semantic Analysis

- $X \sim U_{V \times k} W_{k \times k} V^T_{k \times D}$
- The j th column of WV^T is the representation of a specific document of size $k \times 1$.
- k is the number of latent dimensions
- Large magnitude in dimension $i \rightarrow$ document strongly associated with topic i .
- Use top words and specific documents for interpretation.

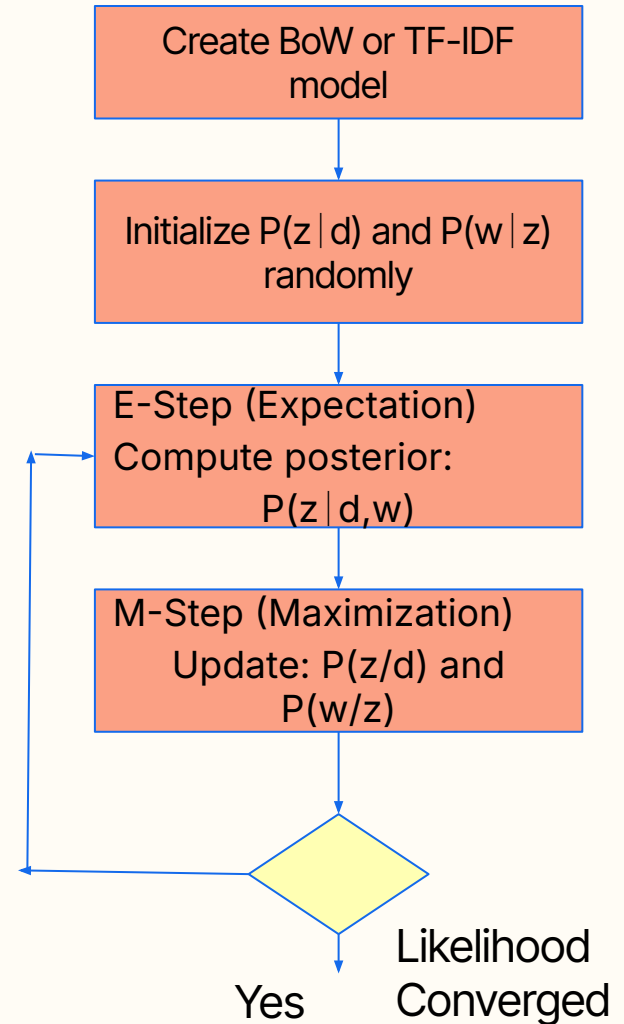


Probabilistic View

- A document is a mixture of topics
- We assume documents are generated by a hidden probabilistic process.
- Assignment of topics is like learning a latent probability distribution from data.
- Eg. The new smartphone features a high-performance processor and a camera optimized for recipe photography.
- Technology:0.75, cooking:0.20, politics: 0.05

Probabilistic LSA

- $P(z/d) = P(\text{topic} | \text{document})$,
 $P(\text{word} | \text{topic}) = P(w/z)$
- Log-Likelihood of the corpus is $\log(P(w/d))$:
- $L = \sum \sum n(d,w) \log(\sum P(w | z) P(z | d))$
- $n(d,w)$ = count of word w in document d
- z = latent topic
- $P(w/d) = \log(\sum P(w | z) P(z | d))$



Latent Dirichlet Allocation

- Disadvantage of pLSA: Number of parameters grow with number of documents
- $P(z | d)$ is learned separately for every document
- Instead of directly learning $P(z | d)$, assume: $\theta_d \sim \text{Dir}(\alpha)$

Latent Dirichlet Allocation

- So document-topic distributions are drawn from a prior: Dirichlet Distribution.
- pLSA + Bayesian prior = LDA
- Methods such as variational EM or Gibbs Sampling are used for approximate inference.